

$$x - y = 3$$

$$y - z = 5$$

$$x - z = 8$$

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 8 \end{pmatrix}$$

equations
↓
redundant

$$(x - y) + (y - z) = x - z$$

$$3 + 5 = 8$$

Only have 2 equations
3rd redundant

2 eqns 3 var $\Rightarrow$ Expect ∞ sol

$$x = 3 + y = 8 + z$$

$$y - z = 5 \quad y = 5 + z$$

$$(3 + y) - z = 8 \Leftrightarrow y - z = 5$$

z is a free parameter

1. $\frac{1}{x^2} = x^{-2}$
 $\frac{d}{dx} x^{-2} = -2x^{-3} = -\frac{2}{x^3}$
 $\frac{d}{dx} \frac{1}{x^2} = -\frac{2}{x^3}$

2. $\frac{d}{dx} \ln(x) = \frac{1}{x}$
 $\frac{d}{dx} \ln(x^2) = \frac{1}{x^2} \cdot 2x = \frac{2}{x}$

3. $\frac{d}{dx} e^x = e^x$
 $\frac{d}{dx} e^{2x} = e^{2x} \cdot 2 = 2e^{2x}$

4. $\frac{d}{dx} \sin(x) = \cos(x)$
 $\frac{d}{dx} \cos(x) = -\sin(x)$
 $\frac{d}{dx} \tan(x) = \sec^2(x)$
 $\frac{d}{dx} \cot(x) = -\csc^2(x)$

$$x + z = 3$$

$$y + z = 5$$

$$x + y = 8$$

Solve

$$x = 3$$

$$y = 5$$

$$z = 0$$

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 8 \end{pmatrix}$$

$$x + y = 1 \cdot x + 1 \cdot y + 0 \cdot z$$

$$x + y - 3 = 0$$

$$x + y = 3$$

$$x + z = 3$$

$$y + z = 5$$

$$x - y = -2$$

$$x + y = 8$$

$$2x = 6$$

$$0.73x + 0.68y + 0.23z = 0.66$$

$$0.07x + 0.90y + 0.89z = 0.88$$

$$0.49x + 0.55y + 0.14z = 0.07$$

Solve for x, y, z

- Could say no sol.

- Could say unique sol

$$x = 2.63$$

$$y = -3.27$$

$$z = 4.10$$

Expect this case

because constraints = equations

Same as variables

$m = n$ Square Matrix

1. The first part of the paper is devoted to the study of the properties of the function $f(x)$ defined by the equation

$$f(x) = \int_0^x \frac{1}{1+t^2} dt$$

and to the investigation of its behavior as $x \rightarrow \infty$.

In the second part we shall consider the function

$$g(x) = \int_0^x \frac{t}{1+t^2} dt$$

$$h(x) = \int_0^x \frac{t^2}{1+t^2} dt$$

and shall study their properties and the limits of these functions as $x \rightarrow \infty$.

Finally, in the third part we shall consider the function

$$k(x) = \int_0^x \frac{t^3}{1+t^2} dt$$

and shall study its properties and the limits of this function as $x \rightarrow \infty$.

$$X = \begin{bmatrix} 0.73 & 0.68 & 0.23 \\ 0.07 & 0.90 & 0.89 \\ 0.49 & 0.55 & 0.14 \end{bmatrix}$$

$$V = \begin{bmatrix} 0.66 \\ 0.88 \\ 0.07 \end{bmatrix}$$

Matrix
2 indices
rows + cols

Vector
1 index

Vector is not a matrix

Can be reshaped to matrix

a Column vector

1 column

$$\begin{bmatrix} 0.66 \end{bmatrix}$$

or a Row vector

1 row

$$\begin{bmatrix} 0.88 \end{bmatrix}$$

$$\begin{bmatrix} 0.07 \end{bmatrix}$$

$$\begin{bmatrix} 0.66 & 0.88 & 0.07 \end{bmatrix}$$

$$\begin{bmatrix} 0.66, 0.88, 0.07 \end{bmatrix} \quad \begin{bmatrix} \begin{bmatrix} 0.66, 0.88, 0.07 \end{bmatrix} \end{bmatrix}$$

THESE ARE THE RESULTS

OF THE INVESTIGATION

CONDUCTED BY THE

COMMISSIONER OF THE

LAND OFFICE

[REDACTED]

[REDACTED]